

ZS-A31

The Effective One-Parameter Reduction of the Z-Spin Vacuum-Energy Line: A Predictivity Consolidation of the Effective Scale M_{eff} , and the Honest No-Go Boundary of the Cosmological Scale

Author: Kenny Kang

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Theme / Code: Astrophysics / ZS-A31 · Consolidation of ZS-F33–F36 · Downstream of ZS-M44/M45

Verification: 26/26 PASS | **Zero fitted parameters in the derived sector** | modulo the register-trace-normalization condition (ZS-M44 v1.5, **DERIVED-CONDITIONAL**), the ZS-F33–F36

line reduces to **one effective dimensional parameter** $M_{eff} := C_{UV}^{1/4} M_K$; the ratio

$\rho_{\Lambda Z}/M_{eff}^4 = \frac{1}{2}(1260/4807)\omega^2$ is the **DERIVED structural number**; the physical scales M_K, ℓ, C_{UV} are **NON-CLAIM / OPEN**, and observation fixes only $M_{eff} \approx 2.48$ meV behind the derivation $\perp$ regression firewall | (A, Q, dim Z) = (35/437, 11, 2) **LOCKED**.

§0. Abstract

ZS-M44 v1.5 (integrating ZS-M45) provides a **graph-Laplacian coordinate diagnostic** whose sector Rayleigh quotients differ; this is **not** the genuine Regge Hessian $B^\top W_{hinge} B$, which remains **OPEN**, so the residual is the action-level selection of a register measure ($\kappa^2 = A/Q$ DERIVED-CONDITIONAL). ZS-F36 v2.1 terminates at $\chi_-^{(s)} = (1260/4807)C_{UV}M_K^4$ with C_{UV} OPEN. Modulo those conditionals, the ZS-F33–F36 line reduces to **one effective dimensional parameter** — the odd-form condensate scale M_{UV} (more precisely $M_{eff} := C_{UV}^{1/4} M_K$). This paper *consolidates* that accounting; it is a predictivity-accounting paper, **not** a derivation of M_{UV} .

Theorem A31.1 (Effective One-Parameter Reduction). Every factor relating the Z-Spin dark-energy density to the **effective scale** $M_{eff} := C_{UV}^{1/4} M_K$ is dimensionless: from ZS-F36 v2.1,

$$\chi_-^{(s)} = (1260/4807)C_{UV}M_K^4 = (1260/4807)M_{eff}^4 \text{ with } 1260/4807 = (dimY)^2 A/Q \text{ (M44 v1.5,}$$

DERIVED-CONDITIONAL on register-trace normalization), and from ZS-F33, $\rho_{\Lambda Z} = \frac{1}{2}\chi_- \omega^2$. Hence

$$\frac{\rho_{\Lambda Z}}{M_{eff}^4} = \frac{1}{2} \cdot \frac{1260}{4807} \cdot \omega^2 = 0.668952 \dots \quad [DERIVED \text{ structural number; } M_{eff} := C_{UV}^{1/4} M_K].$$

Consolidation (behind the firewall). Fixing M_{eff} from the single (Λ CDM-based) observation

$\rho_\Lambda \approx (2.24, meV)^4$ gives $M_{eff} \approx 2.48$ meV. **Only M_{eff} is fixed**; the physical Kähler scale M_K , the length $\ell = M_K^{-1}$, and C_{UV} are **not** — since $M_{eff} = C_{UV}^{1/4} M_K$. Only if $C_{UV} = 1$ (register-tree canonical branch) does $\ell \approx 80, \mu m$ follow. This is **regression**, kept strictly separate from (0.1).

No-go boundary (honest). the *scale* M_{UV} is **not derivable** from dimensionless data: ZS-A17 (Spin–Metric Independence) and ZS-A27/A28 (the A–Q–Only Scale-Generation No-Go, Buckingham- π) forbid it. The **ratio** M_{UV}/M_p^- is a *different* question — an **OPEN debt** (ZS-A22 barrier B3), **not** forbidden (it is dimensionless; §6). The ZS-M3/F10 timescale hierarchy $\tau_n = t_p \exp(n\pi/A)$ *spans* the range but does not fix H_0 at integer n ($n_{Hubble} \approx 3.57$). A31 therefore **consolidates** M_{UV} as the single dimensional anchor and does **not** close the H_0/M_p debt. §6 deep-explores that debt (B3): it reduces to the single number $M_{UV}/M_p \sim 2 \times 10^{-31}$, is shown to be a *debt not a no-go*, and yields one falsifiable HYPOTHESIS-weak (A^{28}); the verdict is **genuinely OPEN** — closure needs the ZS-A17 emergent-metric advance, beyond current corpus tools. Verification 12/12 PASS.

Epistemic Status Legend

Tag	Meaning
DERIVED	Follows from PROVEN corpus results; no new parameter (the dimensionless sector).
REGRESSION	Fixed from observation, behind the derivation $\perp$ regression firewall (the one dimensional input).
NON-CLAIM	Explicitly <i>not</i> asserted (here: a derivation of M_{UV}).
NO-GO	A proven impossibility (ZS-A17/A27/A28) disciplining the program.
OPEN	A genuine remaining debt (here: H_0/M_p).
HYPOTHESIS-weak	A pre-registered, falsifiable numerical relation lacking a derived mechanism (anti-numerology pending).
IDENTITY	Arithmetic consistency.

§1. Introduction: what is, and is not, left

The ZS-F33–F36 line reduced the Z-Spin cosmological constant to $\rho_{\Lambda Z} = \frac{1}{2}\chi_- \omega^2$ with $\chi_-^{(s)} = (1260/4807)C_{norm} M_{UV}^4$, and (ZS-F36) collapsed the two dimensionless residuals $\tilde{G}_s, c_e$ to one via $C_{norm} = 1$. ZS-M44 v1.5 (integrating ZS-M45) *reduces* the dimensionless factor $1260/4807 = (dimY)^2 A/Q$ to a single **register-trace / Regge-Hessian bridge** ($\kappa^2 = A/Q$ **DERIVED-CONDITIONAL**), **not** yet closed action-independently; and ZS-F36 v2.1 leaves C_{UV} (full 1PI) OPEN. What remains is one *effective* dimensionful number, $M_{eff} = C_{UV}^{1/4} M_K$. This paper states that (conditional) terminus cleanly and bounds it against the corpus’s own no-go theorems. Locked inputs: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, $\omega = 2.2592495540$. **Planck convention (fixed, review 3.2):** the reduced Planck mass $M_p^- = M_p/\sqrt{8\pi}$, from the standard Einstein–Hilbert normalization $S_{EH} = \frac{M_p^2}{2} \int \sqrt{-g} R$. All modular-depth statements (§6.3) use M_p^- unless explicitly noted; the unreduced M_p is flagged where used.

§2. Theorem A31.1: the dimensionless reduction [DERIVED]

The dimensionless content is exact and parameter-free:

$$\chi_-^{(s)} = \frac{1260}{4807} M_{eff}^4 \quad \frac{1260}{4807} = (dimY)^2 \frac{A}{Q} = 36 \cdot \frac{35}{4807} \quad [DERIVED - CONDITIONAL on register - trace norm]$$

$$\rho_{\Lambda Z} = \frac{1}{2} \chi_- \omega^2 \Rightarrow \frac{\rho_{\Lambda Z}}{M_{eff}^4} = \frac{1}{2} \cdot \frac{1260}{4807} \cdot \omega^2 = 0.668952 \dots$$

Every quantity on the right of (2.2) is upstream: 1260/4807 (M44 v1.5, **DERIVED-CONDITIONAL** on register-trace normalization), $\omega = 2.2592 \dots$ (ZS-M1, DERIVED), the factor $\frac{1}{2}$ (ZS-F33 canonical branch). **The framework predicts the pure number** $\rho_{\Lambda Z}/M_{eff}^4$ modulo those conditionals; with the physical Kähler scale, $\rho_{\Lambda Z}/M_K^4 = 0.668952 \cdot C_{UV}$ with C_{UV} OPEN. (*Verified: 1260/4807=36A/Q, $\rho_{\Lambda}/M_{eff}^4=0.668952$.*)

This is the substantive consolidation: the structural number $\frac{1}{2}(1260/4807)\omega^2$ relating $\rho_{\Lambda Z}$ to M_{eff}^4 is DERIVED. **Two caveats keep it from being an unconditional closure:** (i) 1260/4807 is DERIVED-CONDITIONAL on register-trace normalization (M44 v1.5); (ii) $M_{eff} = C_{UV}^{1/4} M_K$ folds in the OPEN full-1PI factor C_{UV} (F36 v2.1 §7). What the framework does **not** predict is any single scale (M_{eff} , M_K , or M_{UV}).

§3. The firewall: fixing M_{eff} by regression [REGRESSION]

The following is regression — one observation fixing one parameter — and is quarantined from §2 by the derivation $\perp$ regression firewall.

Adopting the observed dark-energy density $\rho_{\Lambda} \approx (2.24, meV)^4$ (Planck 2018 Λ CDM) and inverting (2.2):

$$M_{eff} = \left(\frac{\rho_{\Lambda}}{0.668952} \right)^{1/4} \approx 2.48, meV.$$

Observation fixes **only** M_{eff} . The length $\ell = M_K^{-1} \approx 80, \mu m$ follows **only under** $C_{UV} = 1$ (since $M_{eff} = C_{UV}^{1/4} M_K$, so $\ell = C_{UV}^{1/4} / M_{eff}$).

The input ρ_{Λ} (hence $M_{UV} \approx 2.48 meV$) is a **Λ CDM-based cosmological inference**, not a direct measurement; it inherits Λ CDM's model assumptions.

Under $C_{UV} = 1$ this reproduces the ZS-F36 canonical-branch length ($\ell \approx 80, \mu m$) — a consistency of the same single effective input, not a new prediction. (*Verified: $M_{eff} \approx 2.48 meV$; $\ell \approx 79.7 \mu m$ conditional on $C_{UV}=1$.*)

One input, then predictions. Once M_{UV} is fixed by this *one* number, any *second* dimensional observable sharing the odd-form scale becomes a genuine prediction via its DERIVED dimensionless ratio to $\rho_{\Lambda Z}$. Cataloguing such observables is the forward program (§5); this paper does not manufacture them.

§4. The no-go boundary: why M_{UV} is not derivable [NO-GO]

Three independent corpus results forbid deriving M_{UV} internally, and A31 respects all three:

- **ZS-A17 (Spin–Metric Independence, NO-GO).** By Connes’ reconstruction, a 3D metric needs a Dirac operator with Weyl growth $\lambda_n \sim n^{1/3}$. The corpus’s information/spin operators (*i*-tetration transfer, $|\mu| < 1$, spectral dimension $p = 0$; Kraus half-angle $p = 0$; single record-flow inclusion $p = 1$) have no such growth. The 3D metric — hence any metric scale — is an *independent datum*, not reconstructible from A or the spin structure.
- **ZS-A27/A28 (A–Q-Only Scale-Generation No-Go, CLOSED-NEGATIVE).** By Buckingham- π , dimensionless data $(A, Q, topology)$ cannot fix a dimensionful scale. ZS-F36 §8 is the F-sector instance: the Koenigs *shape* $\tau = (\omega + i\kappa_\lambda)/2\pi$ is fixed, but the physical *area* needs $\ell = M_{UV}^{-1}$, an unsupplied length.
- **ZS-A22 (barrier B3, OPEN debt).** No action-level mechanism fixes $H_0/M_p \sim 10^{-61}$. The DERIVED hierarchy $\tau_n = t_p \exp(n\pi/A)$ (ZS-M3 §8 / ZS-F10) spans the range — $n = 5$ gives $\tau_5 \approx 2.6 \times 10^{34}$ yr ($p \rightarrow e^+ \pi^0$ scale) — but the Hubble time lands at $n_{Hubble} = \ln(t_H/t_p)/(\pi/A) \approx 3.57$, **not** an integer. So the hierarchy does not pin H_0 ; B3 remains OPEN. (*Verified: $\pi/A \approx 39.22$, $n_{Hubble} \approx 3.57$ non-integer.*) A31 makes **no** attempt to circumvent these. Fixing M_{UV} is regression (§3), not derivation.

§5. Predictivity ledger and NON-CLAIM

Ledger.

Quantity	Status
$1260/4807 = (dimY)^2 A/Q$	DERIVED (M44)
$\rho_{\Lambda,Z}/M_{eff}^4 = \frac{1}{2}(1260/4807)\omega^2 = 0.669$	DERIVED
$\kappa^2 = A/Q$ (register-trace normalization)	DERIVED-CONDITIONAL (M44 v1.5)
$M_{eff} \approx 2.48$ meV	REGRESSION (one observation)
$\ell \approx 80, \mu\text{m}, M_K, C_{UV}$ individually	OPEN (fixed only if $C_{UV} = 1$)
M_{UV} derived internally	NON-CLAIM (A17/A27/A28 NO-GO)
$H_0/M_p \sim 10^{-61}$ from an action mechanism	OPEN (A22 B3; §8, genuine)
B3 reduces to M_{UV}/M_p^- ; a debt not a no-go (separated from the scale no-go)	DERIVED / 60% (§6.1–6.2)
$M_K/M_p^- = e^{-2\pi Q}$ (modular depth; 2π BW-forced; $M_{eff} = C_{UV}^{1/4} M_K$)	HYPOTHESIS-weak / 30% (§6.3; needs M46/M47/F37)
$M_{UV}/M_p \approx A^{28}$	HYPOTHESIS-weak / 30%, superseded by (8.3)

NON-CLAIM (explicit). This paper does **not** derive M_{UV} , does not claim the $\sim$ meV coincidence of the dark-energy and neutrino scales is explained, and does not close the H_0/M_p hierarchy. Its content is the

reduction to one parameter and the *bounding* of that parameter by the corpus's own no-go theorems — an honest predictivity statement, not a solution to the cosmological constant problem.

Forward program. The single external advance that would change the verdict is an emergent-metric proof (a metric-dimension-3 Dirac operator plus a genuine one-sided inclusion), which ZS-A17 names as the missing datum. Absent that, M_{UV} stays the one dimensional anchor.

§6. Deep exploration of the last OPEN item: the B3 hierarchy

The single genuine OPEN in the ZS-F33–F36 line is ZS-A22 barrier **B3**: no action-level mechanism fixes $H_0/M_p \sim 10^{-61}$. This section deep-explores it. The honest terminus is reached in advance: **B3 stays OPEN** — but the exploration sharpens it materially.

§6.1 Reduction to one number [DERIVED]

H_0/M_p and ρ_Λ/M_p^4 are not independent unknowns. By the Friedmann relation $H_0^2 M_p^2 \sim \rho_\Lambda$,
 $H_0/M_p \sim (\rho_\Lambda/M_p^4)^{1/2} \sim (M_{UV}/M_p)^2 \times O(1)$ (standard, not new); and $\rho_\Lambda/M_p^4 = (M_{UV}/M_p)^4 \times 0.669$ (§2, DERIVED). The **observed cosmological hierarchy reduces to the effective ratio M_{eff}/M_p^-** ; its *mechanism*, however, does **not** reduce to a single microscopic quantity, because $M_{eff} = C_{UV}^{1/4} M_K$ with C_{UV} and M_K separately unresolved. So **B3 reduces to one observed number**:

$$\frac{M_{eff}}{M_p^-} \approx \frac{2.48, meV}{2.435 \times 10^{30}, meV} = 1.02 \times 10^{-30} \quad (\log_{10} = -29.99),$$

using the **reduced** $M_p^- = M_p/\sqrt{8\pi}$ fixed in Locked Inputs (the unreduced M_p would give 2.0×10^{-31} ; the convention must be fixed *before* any modular-depth comparison, §6.3).

Everything else in the cosmological hierarchy follows from this one ratio. (*Verified: B3 reduces to one number; $H_0/M_p \sim (M_{UV}/M_p)^2$.*) **[90%]** that this is the correct reduction.

§6.2 B3 is a debt, not a no-go [60%]

A crucial distinction (sharpened per review 3.1): **two different questions must not be conflated.** (a) Deriving the *scale* M_{UV} from A, Q alone is a **NO-GO** (ZS-A27/A28 Buckingham- π ; ZS-A17 metric independence). (b) Deriving the *dimensionless ratio* M_{UV}/M_p^- from **second-scale dynamics** (a mechanism relating two scales) is an **OPEN debt**, not forbidden. The scale no-go applies to (a), not (b). What it needs is a *second* scale (M_p) related to M_{UV} by a **derived mechanism**. The corpus has no such mechanism (that absence *is* B3). So B3 is properly classified not as an impossibility (unlike the ZS-A17 metric no-go) but as an **open debt with a concrete, finite target** (8.1). This demotes B3 from “possibly forbidden” to “open, and in-principle closable by a mechanism.” (*Verified: ratio dimensionless => debt not no-go.*)

§6.3 Candidate structures and anti-numerology

The τ_n hierarchy misses. The DERIVED timescale hierarchy $\tau_n = t_p \exp(n\pi/A)$ (ZS-M3/F10) gives $M_{UV}/M_p = \exp(-n\pi/A)$ at $n = 1.80$ — **non-integer** (the nearest integer $n = 2$, the weak scale, misses (8.1) by $\exp(0.2\pi/A) \approx 2400 \times$). The hierarchy does not select the dark-energy scale. (*Verified: $n=1.80$ non-integer.*)

The primary mechanism frontier: modular depth $e^{-2\pi t_*}$ **[HYPOTHESIS-weak, promoted]**. A half-sided modular inclusion $N \subset M$ (Borchers–Wiesbrock) carries an *intrinsic* scale action: the positive generator $P = \frac{1}{2\pi}(\log \Delta_N - \log \Delta_M) \geq 0$ satisfies

$$\Delta_M^{it}, P, \Delta_M^{-it} = e^{-2\pi t}, P.$$

The 2π here is **not a fit** — it is forced by the modular (Tomita–Takesaki / Borchers–Wiesbrock) structure [Borchers 1992; Wiesbrock 1993]. (The inclusion orientation, modular-flow sign, and $P \geq 0$ convention must match the standard +half-sided form; a genuine — non-conditional-expectation — inclusion is required, §6.4.) If a *genuine* one-sided modular inclusion and a metric-dimension-3 modular Dirac operator can be constructed (the ZS-A17 advance), then an emergent mass scale evaluated at modular depth t_* obeys $M_{UV}/M_P^- = e^{-2\pi t_*}$ — an **action-level** hierarchy. This is a materially better frontier than any A^k fit, because the exponent's 2π is structurally mandated.

Numerically, with the fixed reduced convention $M_P^-: t_{obs} = -\frac{1}{2\pi} \log(M_{UV}/M_P^-) \approx 10.991$, i.e.

$t_{obs} - Q \approx -0.0089$. With the unreduced $M_P^-: t_{obs} \approx 11.25 \approx Q + \frac{1}{4}$. **Convention sensitivity is**

decisive (review 3.2): the two conventions select Q vs $Q + \frac{1}{4}$, so choosing post-hoc would be numerology. Having fixed M_P^- in Locked Inputs, the **single pre-registered candidate** is

$$\frac{M_K}{M_P^-} \stackrel{?}{=} e^{-2\pi Q} \quad [\text{HYPOTHESIS} - \text{weak}, 30]$$

Which scale does the candidate predict? (review C). The modular mechanism would fix the *metric/Kähler* scale M_K , not the effective scale. So (6.3) is the **emergent-metric candidate**

$M_K/M_P^- \stackrel{?}{=} e^{-2\pi Q}$. The **observed** ratio is the effective one, $M_{eff}/M_P^- = 1.018 \times 10^{-30}$, related by

$M_{eff} = C_{UV}^{1/4} M_K$, so

$$\frac{M_{eff}}{M_P^-} = C_{UV}^{1/4} e^{-2\pi Q}.$$

Numerically $e^{-2\pi Q} = e^{-22\pi} = 9.632 \times 10^{-31}$, so the observed 1.018×10^{-30} implies $C_{UV}^{1/4} \approx 1.057$, i.e. $C_{UV} \approx 1.25$. **Therefore the ~ 5.4 gap is undecidable until the full 1PI is computed:** it is *either* a falsification of the exact modular relation *or* exactly the UV threshold $C_{UV} \approx 1.25$ (F36 v2.0 §7, OPEN).

The pre-registration is that the *action* yields $M_K/M_P^- = e^{-2\pi Q}$ **exactly** and, independently, that the 1PI yields C_{UV} ; the 5.4 must **not** be absorbed post-hoc into Q (no $Q \rightarrow Q - 0.0089$) — only a *computed* C_{UV} may account for it. The phenomenological alternative $M_{eff}/M_P^- \stackrel{?}{=} e^{-2\pi Q}$ is a *direct* 5.4 mismatch and is **not** the pre-registered candidate.

The unreduced $Q + \frac{1}{4}$ form is admissible **only** if a Z-sector quarter-step ($\Delta_Z^{1/4}$ seam operator) is separately *proven* from the action. Closing (8.3) to DERIVED needs three theorems, none yet in the corpus: a genuine HSMI (proposed **ZS-M46**), a metric-dimension-3 modular Dirac operator ($N_D(\Lambda) \sim C\Lambda^3$; proposed **ZS-M47**), and an action that *selects* $t_* = Q$ via a Q -unit inclusion chain

$M_0 \supset \dots \supset M_Q$ (proposed **F37/A32**). Until then, (8.3) is a frontier hypothesis, not a result. (*Verified: $t_{\text{obs}}(\text{reduced}) \sim Q$, $t_{\text{obs}}(\text{unreduced}) \sim Q + 1/4$, 2π BW-forced, $HSMI + p3 + \text{depth}$ unproven.*)

A demoted coincidence: A^{28} [HYPOTHESIS-weak, superseded by (8.3)]. For completeness (now secondary to the modular frontier): the exponent solving $M_{UV}/M_P = A^k$ is $k = 27.99$, strikingly close to the structural integer $|V_Y - F_Y| = 60 - 32 = 28$ (the truncated-icosahedron vertex–face asymmetry, numerator of $\delta_Y = 28/92 = 7/23$). Numerically $A^{28} = 2.00 \times 10^{-31}$ versus 2.03×10^{-31} (1.7%).

An anti-numerology scan finds **no other** structural formula

$c^k : c \in A, |\lambda^*|, \kappa_\lambda, 1/\omega, k \in \text{corpus integers}$ within 2% of (8.1); the random chance rate of hitting the ± 2 band is $p \approx 0.005$ (**exploratory only, review 3.3: the “structural-formula universe” was not pre-registered, so no multiple-testing penalty is computable — this p is not a valid significance and should not be cited as one**).

Honest anti-numerology caveats (why this is only HYPOTHESIS-weak, not DERIVED):

1. **Nearest-integer, not forced.** The band $[A^{28.5}, A^{27.5}]$ spans a factor ~ 12 ; any M_{UV}/M_P within a factor ~ 3.5 of the true value would equally “select” $k = 28$. The content is only that 27.99 rounds to a structural 28 — not that 28 is derived.
2. **No mechanism.** A^{28} is a *number*. B3 asks for an *action-level mechanism*. The ZS-M3/M45 holonomy accumulates a *phase* $\delta\phi = A$ per cell, not a *suppression* A ; there is no derived instanton/tunneling process producing A^{28} (a 28-fold Y-sector suppression would be needed, and is not in the corpus).
3. **Post-regression.** M_{UV} is itself fixed by regression to ρ_Λ (§3), so (8.1) is an observed input; matching it is post-diction.
4. **The §10.4 lesson.** ZS-M44 §5.4 caught an analogous trap — a numerical ratio (0.855) spuriously mimicking a structural constant (δ_X/δ_Y) — that dissolved under exact treatment. A^{28} carries the same warning: a structural-looking exponent is not a derivation.

Falsification: if M_{UV}/M_P is pinned more precisely (better ρ_Λ) and departs from A^{28} beyond the exponent-rounding tolerance, the hypothesis dies. It is registered **only** for tracking, at **30%**, and is **superseded as the primary frontier by the modular-depth candidate (8.3)**, whose exponent 2π is structurally forced rather than a nearest-integer fit. (*Verified: A^{28} within 2%, $k=27.99$ nearest-int, 0 other formulas within 2%, no mechanism.*)

§6.4 The mechanism gap and the emergent metric [OPEN]

Closing B3 to DERIVED requires an action-level mechanism relating M_{UV} to M_P . The only candidate direction — a genuine H_0 requires a real 3D metric — is blocked by ZS-A17 (Spin–Metric Independence NO-GO): the corpus’s information/spin operators are metric-dimension 0–1, so no emergent metric-dimension-3 Dirac operator exists in the current toolset. The single external advance ZS-A17

names — a metric-dimension-3 Dirac operator plus a genuine one-sided modular inclusion — is **beyond current corpus tools**, but is now **concretely scoped** by §6.3: ZS-M46 (genuine HSML, e.g. a Berry–Keating standard pair), ZS-M47 ($D_3 = \gamma^i P_i$ from three compatible inclusions, $N_D(\Lambda) \sim C\Lambda^3$), and an action selecting $t_* = Q$. No mechanism is *found* here, but the path is a defined program rather than a void. (*Verified: emergent-metric = single advance, beyond tools.*)

§6.5 Verdict [OPEN, genuine]

B3 stays OPEN — confirmed. Per the deep-exploration protocol, an exploration that converges with its terminal node OPEN is a *genuine OPEN*: the result is the confirmation that “current corpus tools cannot close it.” The exploration’s value is the sharpening: B3 is one number (8.1), a **debt not a no-go** (§6.2, cleanly separated from the scale no-go), with a **structurally-motivated** frontier hypothesis

$$M_K/M_P^- = e^{-2\pi Q} \quad (\S 6.3; \text{the } 2\pi \text{ Borchers–Wiesbrock-forced; the observed } 5.4 \text{ gap} \Leftrightarrow C_{UV} \approx 1.25,$$

undecidable pre-1PI) superseding the A^{28} nearest-integer fit, and a **concrete program** (ZS-M46/M47/F37) to close it. **[90%]** B3 remains OPEN under current corpus tools.

Deep-exploration record. *Step 0 (long list)*: (1) the target number; (2) τ_n hierarchy; (3) A^k ; (4) does

A27/A28 forbid the ratio; (5) an action mechanism; (6) emergent metric; (7) anti-numerology. *Step 1 (MECE)*: I1 target-reduction {1}; I2 no-go-vs-debt {4}; I3 candidates+anti-numerology {2,3,7}; I4 mechanism+metric {5,6}. *Step 3 (status)*: I1 DERIVED (8.1); I2 DEBT-not-NO-GO (§6.2, 60%); I3 τ_n

misses, A^{28} HYPOTHESIS-weak (30%); I4 OPEN (no mechanism; emergent metric beyond tools). *Step 4 (convergence)*: change counts [4, 1, 0] decreasing $\Rightarrow$ converged with terminal node I4 OPEN $\Rightarrow$ genuine OPEN. *Step 5 (value)*: not a closure, but a sharpening — one number, a debt-reclassification, a falsifiable hypothesis, and a named advance; anti-numerology enforced (the A^{28} trap explicitly flagged, per the §10.4 lesson).

§7. Falsification, anti-numerology, cross-version

Falsification. *F-A31.1*: if $1260/4807 \neq (\dim Y)^2 A/Q$ or the M44 chain reopened, (2.2) fails. *PASS*. *F-A31.2*: if the τ_n hierarchy fixed H_0 at integer n , B3 would close and part of the no-go would lift. *PASS* ($n \approx 3.57$, non-integer). *F-A31.3*: a claim to derive M_{UV} from (A, Q) alone is void by A27/A28.

Anti-numerology. The only *derived* number is the dimensionless $0.669 = \frac{1}{2}(1260/4807)\omega^2$; it is a consequence of M44 + F33, not a fit. The dimensional $M_{eff} \approx 2.48 \text{ meV}$ is explicitly REGRESSION (one observation), not a coincidence claim. Anti-numerology Monte Carlo is **not applicable to Theorem A31.1 itself**. The numerical candidates in §6.3 remain **exploratory** because their formula space was not preregistered; no confirmatory significance is claimed.

Cross-version. ZS-F33/F34/F35/F36, ZS-M44/M45, ZS-A17/A22/A27/A28: no value modified. A31 consolidates; it changes no upstream status. $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

§8. Verification, references, version history

zs_a31_b3_verify.py, **12/12 PASS** (extends the v1.0 6/6): the v1.0 core plus the §6 B3 exploration — $1260/4807 = 36A/Q$; $\rho_{\Lambda Z}/M_{eff}^4 = 0.668952$ (dimensionless structural number,

DERIVED-CONDITIONAL); $M_{eff} \approx 2.48$ meV (regression), $\ell \approx 79.7$ μm conditional on $C_{UV} = 1$; $\pi/A \approx 39.22$; $n_{Hubble} \approx 3.57$ (non-integer, B3 OPEN); and the consolidation/no-go flag; and the §6 checks — B3 reduces to one number M_{UV}/M_P ; τ_n misses ($n = 1.80$); $|V_Y - F_Y| = 28$; A^{28} within 2% with $k = 27.99$ nearest-integer and 0 other structural formulas within 2%; the debt-not-no-go and no-mechanism flags; the emergent-metric requirement.

References. [1] ZS-F33/F34/F35/F36 v2.1 (Z-Spin corpus). [2] ZS-M44 v1.5 (integrating ZS-M45 v1.1) (Z-Spin corpus). [3] ZS-A17 (Spin–Metric Independence NO-GO), ZS-A22 (barrier B3), ZS-A27/A28 (Scale-Generation No-Go) (Z-Spin corpus). [4] ZS-M3 §8 / ZS-F10 (timescale hierarchy $\tau_n = t_p e^{n\pi/A}$).

[5] Planck Collaboration, *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.

Added references (v1.3). H.-J. Borchers, *The CPT theorem in two-dimensional theories of local observables*, *Commun. Math. Phys.* **143**, 315 (1992). H.-W. Wiesbrock, *Half-sided modular inclusions of von Neumann algebras*, *Commun. Math. Phys.* **157**, 83 (1993). ZS-M45 v1.1 is **integrated into ZS-M44 v1.5 §6** (cite M44 v1.5).

Version History. *v1.5 (July 2026, sync-corrected — terminal)*: Synchronizes A31 with the finalized companions **ZS-M44 v1.5** and **ZS-F36 v2.1** and integrates the four review items: (3.1) the §0 abstract now states M44 v1.5’s honest terminus — a **graph-Laplacian coordinate diagnostic**, *not* the genuine Regge Hessian $B^T W_{hinge} B$ (OPEN), the residual being the action-level register-measure selection; (3.2) §2 eqs (2.1)–(2.2) use the **effective scale** M_{eff} and tag the 1260/4807 factor

DERIVED-CONDITIONAL on register-trace normalization (was “M44-DERIVED”); (3.3) the §3 title and the §6.1 hierarchy are reframed so that the **observed** ratio M_{eff}/M_P^- reduces to one number while its *mechanism* (C_{UV}, M_K) stays OPEN — removing the “one number” vs “both OPEN” tension; (3.4) the §7 anti-numerology paragraph is corrected (MC not applicable to Theorem A31.1; the §6.3 candidates are exploratory, non-preregistered). Reference updates M44 v1.3→v1.5, F36 v2.0→v2.1. No numerical value changed; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. *v1.4 (July 2026, third-review-corrected)*: **Decisive conceptual fix**: the single observation fixes only the **effective scale** $M_{eff} = C_{UV}^{1/4} M_K \approx 2.48$ meV — not M_K , ℓ , or C_{UV} ; so $\ell \approx 80$ μm is conditional on $C_{UV} = 1$. Title → “Effective One-Parameter Reduction.” All stale closure language removed (banner “dimensionless chain complete”; intro “M44 discharged the last caveat / M45 realized it”; ledger “all M6 caveats discharged / $\kappa^2 = A/Q$ DERIVED”; verification “ ρ/M_{UV}^4 DERIVED”) → all now DERIVED-CONDITIONAL, consistent with F36 v2.0 / M44 v1.5. **Modular candidate pre-registered on the metric scale**

$M_K/M_P^- = ? e^{-2\pi Q}$ (emergent-metric reading), with $M_{eff}/M_P^- = C_{UV}^{1/4} e^{-2\pi Q}$ so the 5.4 gap $\Leftrightarrow C_{UV} \approx 1.25$ — undecidable until the full 1PI; the phenomenological M_{eff} reading (direct 5.4 mismatch) is explicitly *not* the candidate. Anti-numerology paragraph sharpened. References → ZS-M44 v1.5, F36 v2.0. B3 remains **OPEN**; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. *v1.3 (July 2026, second-review-corrected)*: Synchronizes with **F36 v1.9 / M44 v1.3**: the dimensionless normalization is

DERIVED-CONDITIONAL (register-trace) and folds in the OPEN full-1PI factor C_{UV} , so (0.1) is stated via $M_{eff} = C_{UV}^{1/4} M_K$ and the physical form carries C_{UV} OPEN; M44/M45 are cited as *reducing* (not

closing) the problem to a register-trace/Regge-Hessian bridge. **Fixes eq (8.1) to the reduced M_p^-** (

$M_{UV}/M_p^- \approx 1.02 \times 10^{-30}$, $\log_{10} \approx -29.99$), consistent with Locked Inputs. **States the modular**

candidate's honest 5.4% gap ($e^{-2\pi Q} = 9.63 \times 10^{-31}$ vs regression 1.02×10^{-30} ; ratio 0.946) and pre-registers it as an *exact* target, to be falsified if the action gives no independent correction. Adds Borchers/Wiesbrock references and the HSMI convention note. Reconciles the §7 anti-numerology paragraph with the §6.3 exploratory hypotheses. Reorders §8→§6 (B3 exploration now follows the ledger). Notes $M_{UV} \approx 2.48$ meV is a Λ CDM-based inference. B3 remains **OPEN**;

($A, Q, \dim Z$) = (35/437, 11, 2) LOCKED. *v1.2 (July 2026, review-corrected)*: Integrates external review. (1) Separates the **no-go** (scale M_{UV} from A, Q) from the **debt** (ratio M_{UV}/M_p^- from second-scale dynamics), removing the abstract/§8 conflict (review 3.1). (2) **Fixes the Planck convention** to reduced M_p^- in Locked Inputs (review 3.2). (3) **Promotes the modular-depth frontier** $M_{UV}/M_p^- = e^{-2\pi t_*}$ with the 2π Borchers–Wiesbrock-forced, pre-registering the single candidate $e^{-2\pi Q}$ (reduced convention) and scoping the closing program ZS-M46/M47/F37; **demotes** A^{28} (nearest-integer fit) to secondary. (4)

Relabels the A^{28} Monte-Carlo $p = 0.005$ as exploratory-only (no pre-registered formula universe, review 3.3). B3 remains **OPEN**; ($A, Q, \dim Z$) = (35/437, 11, 2) LOCKED. *v1.1 (July 2026)*: Integrates a deep exploration of the last OPEN item, ZS-A22 barrier **B3** (§8). Verdict: **B3 stays OPEN** (genuine — no action-level mechanism in the current corpus). Sharpened: (i) B3 reduces to the single number

$M_{UV}/M_p^- \approx 2.0 \times 10^{-31}$ (§6.1); (ii) it is a **debt, not a no-go** — the ratio is dimensionless, so

ZS-A27/A28 do not strictly forbid it (§6.2, 60%); (iii) the coincidence $M_{UV}/M_p^- \approx A^{|V_Y - F_Y|} = A^{28}$ (1.7%,

$k = 27.99$, unique within 2%) is registered **HYPOTHESIS-weak** (30%) with explicit anti-numerology caveats (nearest-integer, no mechanism, post-regression, the §10.4 trap lesson); (iv) closure needs the ZS-A17 emergent-metric advance, beyond current tools. 12/12 checks; no numerical value changes;

($A, Q, \dim Z$) = (35/437, 11, 2) LOCKED. *v1.0 (July 2026)*: Initial release. Consolidates the ZS-F33–F36 vacuum-energy line to one free dimensional parameter M_{UV} : the dimensionless ratio

$\rho_{\Lambda,Z}/M_{UV}^4 = \frac{1}{2}(1260/4807)\omega^2 = 0.669$ is DERIVED (Theorem A31.1); $M_{UV} \approx 2.48$ meV ($\ell \approx 80, \mu$ m) is fixed by one observation behind the firewall (REGRESSION); and M_{UV} is bounded as

NON-CLAIM / IRREDUCIBLE by ZS-A17/A27/A28, with the H_0/M_p debt (A22 B3) OPEN (

$n_{Hubble} \approx 3.57$). 6/6 checks; zero fitted parameters in the derived sector;

($A, Q, \dim Z$) = (35/437, 11, 2) LOCKED. **Honest scope: this is a predictivity consolidation, not a derivation of M_{UV} nor a solution to the cosmological constant problem.**

Version: v1.5 (supersedes v1.4–v1.0; synchronizes with ZS-M44 v1.5 and ZS-F36 v2.1; integrates the four sync-review items — terminal version). **v1.4 correction (decisive):** the single dark-energy

observation fixes only the **effective scale** $M_{eff} := C_{UV}^{1/4} M_K \approx 2.48$ meV — **not** M_K , ℓ , or C_{UV}

individually. So $\ell \approx 80, \mu$ m is **conditional on** $C_{UV} = 1$ (the register-tree canonical branch). All stale

“dimensionless chain complete / $\kappa^2 = A/Q$ DERIVED / all M6 caveats discharged” language is removed

(F36 v2.0 / M44 v1.5: DERIVED-CONDITIONAL). The modular candidate is pre-registered as $M_K/M_p^- \stackrel{?}{=} e^{-2\pi Q}$ (emergent-metric reading), with the observed $M_{eff}/M_p^- = C_{UV}^{1/4} e^{-2\pi Q}$ so the 5.4 gap $\Leftrightarrow C_{UV} \approx 1.25$ — undecidable (falsification vs UV threshold) until the full 1PI is computed. **v1.2 changes:** (i) cleanly separates the **no-go** (M_{UV} from A, Q alone) from the **debt** (M_{UV}/M_p from second-scale dynamics); (ii) **fixes the Planck-mass convention** (reduced M_p^- , standard Einstein–Hilbert) in Locked Inputs, since M_p vs M_p^- shifts the modular depth by ~ 0.25 ; (iii) demotes the A^{28} coincidence and **promotes the modular-depth frontier** $M_{UV}/M_p = e^{-2\pi t_*}$, where the 2π is **forced by Borchers–Wiesbrock** (not a fit) — a materially better mechanism candidate; (iv) relabels the A^{28} Monte-Carlo p as exploratory.